BIOLOGICAL SCIENCES

SOME ASPECTS OF BIOLOGY AND MOLECULAR GENETIC OF SARS-CoV-2

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Abstract

The purpose of the research was to study the possibility of circulation of the SARS-CoV-2 coronavirus in populations of various species of domestic and wild animals, to study its biological and molecular genetic characteristics. A total of 24 species of animals were involved. Of these, 12 animal species received positive tests for the isolation of virus RNA and the isolation of specific antibodies to SARS-CoV-2: domestic cat, dog, American mink, dark polecat, nasua, Cameroonian goat, pig, donkey, horse, yellow-throated mouse, red vole, budgie. The virus was most often isolated from domestic cats (in 5.59% of samples), American mink (7.51% of samples), and dogs (10.52% of samples). SARS-CoV-2 was isolated from American mink and domestic cat using VERO cell culture, and its full genome sequencing was carried out. During this study, information was obtained about 2 new viruses classified as subtype B.1 according to the Pango classification.

Keywords: SARS-CoV-2, COVID-19, domestic and wild animals, sequencing, mutations.

Formulation of the problem. Despite the fact that the pandemic status of COVID-19 has been canceled, this diseases and the virus itself have not disappeared anywhere. Today this is a problem of humane and veterinary medicine. The virus periodically infects new host species, creating new foci or the disease, which indicates the need to study the issue of susceptibility of various animal species to SARS-CoV-2.

Literature review. The period of late 2019 – mid-2023 is remembered as one of the serious pandemics caused by the coronavirus, named SARS-CoV-2, and manifested by a disease called COVID-19, outbreaks of which continue to be recorded around the world to this day. Given the similarity of the SARS-CoV-2 virus to SARS-CoV, as well as related coronaviruses circulating in bats, it is believed that the SARS-CoV-2 virus formed in certain bat species in China. However, it is still not clear at this time whether the virus was transmitted directly from the bat to humans or through an intermediate host, as well as to wild and domestic animals that may act as reservoirs of the virus. As a result of a number of studies, it has been shown that the SARS-CoV-2 virus is able to replicate in the body of a large range of mammals. This ability of the virus binds to angiotensin-converting enzyme 2 (ACE2), which is a zinc-containing metalloenzyme. It has now been established that ACE2 has an affinity for S-glycoproteins of a number of coronaviruses, including SARS-CoV and SARS-CoV-2 viruses. Given the fact that ACE2 is expressed in most mammalian tissues and it is the basis for the penetration of SARS-CoV-2 into the cell, one can assume a possible risk of infection with this virus in a wide range of domestic animals, livestock and wild animals [1, 2, 3].

During the COVID-19 pandemic, natural cases of transmission of the SARS-CoV-2 virus to animals were reported, including free-roaming white-tailed deer in North America and farmed American mink on several continents [4, 5, 6, 7, 8]. To understand the potential of ACE2-mediated viral tropism, a number of scientists used immunohistochemistry to study the distribution of ACE2 receptors in the tissues of the respiratory tract and intestines of some wild and semi-domesticated mammals, including artiodactyls, species of martens and other mammals. The expression of the ACE2 receptor has been detected in the bronchial or bronchiolar epithelium of several European and Asian species of deer, double-humped camel, European badger, ermine, hippopotamus, sea seal, and hooded seal. Further mapping of ACE2 receptors in the nasal concha and trachea revealed rare expression of the receptor in the epithelial cells of the mucous membrane and the accidental appearance in the submucosal glandular epithelium of Western roe deer, elk and alpaca. Only the European badger and ermine had high levels of the ACE2 receptor in the epithelium of the nasal mucosa, which may indicate a high susceptibility to the causative agent of a new respiratory infection. Expression of the ACE2 receptor in intestinal cells was ubiquitous in many of the taxa studied. The research results demonstrate the potential for ACE2-mediated viral infection in some wild mammals and highlight the internal taxon variability of ACE2 receptor expression, which may affect host susceptibility and infection [1, 2, 4, 9, 10, 11, 12].

Thus, the relevance of the issue of the potential possibility of infection of various animal species and their role in the epidemic process is beyond doubt.

Purpose: The purpose of the research was to study the intensity of circulation of the SARS-CoV-2 virus in populations of domestic and wild animals and to determine the molecular and genetic features of the isolated varieties of the virus.

Materials and Methods: The study of the circulation of the SARS-CoV-2 virus was carried out in populations of domestic and wild animals. A total of 24 species of animals were involved: domestic cat, dog, cattle, sheep and goats, rabbit, guinea pig, chinchilla, domestic pig, horse, donkey, American mink, black fox, European polecat, mongoose, Nasua, red deer, yellow-necked mouse, bank vole, brown rat, budgerigar, bird (chickens, mute swan), rhesus macaque. Biological materials for research were: smears or flushes from the mucous membranes of the oral and nasal cavities (beak in birds), from the mucous membrane of the rectum (cloaca in birds). Pieces of parenchymal organs (lungs, heart with a blood clot, spleen, liver, kidneys, lymph nodes (tonsils of a bird)) were taken from fallen animals. The PCR study was conducted using domestic kits "ArtBioTech" (Minsk, Republic of Belarus) and "CIVital" (Vitebsk, Republic of Belarus). Serological research was conducted using the ID-VET kit (France). Virus isolation was conducted on a VERO cell culture.

Sequencing of samples isolated from animals was performed followed by the assembly of the sequence of the SARS-CoV-2 virus. Nanopore MinION technology with 2600x coating was used. The results were deposited in the Gisaid database [https://gisaid.org /].

Research results. As a result of the monitoring conducted to study the circulation of SARS-CoV-2 in populations of various animal species, we obtained positive research results (isolation of SARS-CoV-2 virus RNA or isolation of specifically antibodies to SARS-CoV-2 virus) in populations of the following 12 animal species (50% of all examined species): domestic cat (Felis catus Linnaeus, 1758), dog (Canis familiaris Linnaeus, 1758), Cameroonian goat (Capra hircus Linnaeus, 1758), domestic pig (Sus domesticus Erxleben, 1777), horse (Equus ferus Boddaert, 1785), donkey (Equus asinus Linnaeus, 1758), American mink (Neovision vison Schreber, 1777), European polecat (Mustela putorius Linnaeus, 1758), nasua (Nasua nasua Linnaeus, 1766), budgerigar (Melopsittacus undulatus Shaw, 1805), yellow-necked mouse (Sylvaemus flavicollis Melchior, 1834), bank vole (Myodes glareolus Schreber, 1780). Specific antibodies were isolated in a domestic cat in 34.9% of blood serum samples, and in a dog in 6.7% of samples, with an antibody titer from 0.705 to 3.361.

In two species of animals – a domestic cat and an American mink, the virus was isolated and sequenced. In the course of this study, information was obtained on 2 new variants of the SARS-CoV-2 virus, belonging to the subtype B.1 according to the Pango classification. Individual mutations in the genetic structure of the virus were also identified. The lists of significant mutations are given in Tables 1 and 2.

Table 1.

The list of distinctive mutations hCoV-19/mink/Belarus/RRPCEM-VIS_2216O/2021. The sample was received from Mustela putorius furo (2020-06-22) – Betacoronavirus Clade GH. Gisaid Accession ID: EPI_ISL_ 2521999.

Gene	Amino Acid substitutions
S	D614G
S	R682Q
N	S194L
NS3	Q57H
NS7a	T61I
NS7a	V93F
NS7b	L6M
NSP2	A360V
NSP8	T141M
NSP12	P323L

Table 2.

The list of distinctive mutations hCoV-19/cat/Belarus/RRPCEM-VIS_1884O/2021. The sample was received from Felis catus (2020/11/19) – Betacoronavirus Clade GH. Gisaid Accession ID: EPI_ISL_2100634.

Gene	Amino Acid substitutions
S	D614G
N	S194L
NS3	Q57H
NS7a	T61I
NS7a	V93F
NS7b	L6M
NSP8	T141M
NSP12	P323L

Discussion Based on numerous studies conducted in different countries, nowadays it has been found that in the early stages of the COVID-19 pandemic (until February 2020), three main groups of mutations were stably fixed in the virus genome, which caused three branches of the evolutionary development (cluster 1-3) of the SARS-CoV-2 virus. The main number of these mutations relates to the ORF1ab gene, but only one of them leads to amino acid substitution, the remaining single mutations are synonymous. But by the end of March 2020, the evolutionary development of the three clusters is slowing down and variants of the SARS-CoV-2 virus with a G mutation in the S gene (position D614G) are mainly detected. It cannot be excluded that variants of the SARS-CoV-2 virus with this mutation appeared already in the first half of January 2020, and then there was a rapid spread of 4 variants of the SARS-CoV-2 virus with this mutation. By April 2020, SARS-

CoV-2 viruses that do not contain the D614G mutation in the genome were found only in 25% of cases. We identified that the G mutation in the S gene (position D614G) in the SARS-CoV-2 virus isolated from animals may indicate that this mutation is associated with the biological properties of the virus, facilitating transmission not only from person to person, but also from person to animals. This feature may be due to the fact that the mutation in the D614G position affects the spike protein of the SARS-CoV-2 virus, which plays an important role in the penetration of the virus into host cells. Mutation in the D614G position is connected with increased infectivity and has become the dominant variant worldwide. Thus, the researchers noted the widespread of SARS-CoV-2 virus strains in people with a mutation in the D614G position, which causes an accelerated process of binding of the virus to the host cell and penetration into it, and the fact that we have established the circulation of such viruses in animals allows us to assume that mutations may occur in the future, leading not only to an even wider spread of the SARS-CoV-2 virus in animals and birds, but also changing its virulence.

The multifunctional structural S-protein of the SARS-CoV-2 virus is necessary to maintain the life cycle of the virus (life style), ensures its adaptation to various conditions of existence, determines its contagiousness and pathogenicity. The SArS-CoV-2 spike protein is of great importance in the pathogenesis of SARS-CoV-2. In addition to tMprSS spike, SarS-CoV-2 can be proteolytically activated by other cellular proteases, such as cathepsin B and l (endosomal cysteine proteases), as well as furin, elastase, factor X and trypsin, which are capable of similar “priming” proteolysis, triggering the process of virus entry into host cells. The R682Q mutation we identified is located in the receptor-binding domain (RBD) of spike protein in the SARS-CoV-2 virus strain isolated from animals, and it can potentially affect the interaction of the virus with the ACE2 receptor, thereby affecting the ability of the virus to penetrate into human and animal cells.

Conclusion Thus, the conducted studies have proved the possibility of circulation of the SARS-CoV-2 virus in the body of various animal species, causing the development of an infectious disease and even death in some species. The full-genome sequencing of the SARS-CoV-2 virus isolated from the body of domestic cat and American mink showed significant mutations in the genetic structure of the pathogen, allowing it to more easily penetrate into the cells of various animal species. The isolation of the virus in large quantities from the body of pets indicates the need for a more in-depth study of this issue in terms of its epidemic and possible epizootic significance.

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